

PROJECT DESIGN PHASE

Project Name: Import Data Using Transform Maps (Spreadsheet)

Module 1: Problem Solution Fit

1. Problem Statement

In enterprise data management, large datasets are frequently accumulated and updated in standalone spreadsheets (.xlsx or .csv). Manually copying and pasting or re-entering this spreadsheet data into centralized database tables creates critical operational roadblocks:

- **High Operational Overhead:** Manual entry demands excessive time and labor.
- **Human Errors & Inconsistencies:** Typographical errors and incorrect field matching degrade data quality.
- **Record Duplication:** Inability to automatically cross-reference existing records leads to duplicate database rows.
- **Risk of Data Corruption:** Direct raw input without a staging layer risks pushing invalid schemas into production tables.

2. Target Audience & Stakeholders

System Administrators, Data Operations Engineers, and Business Analysts responsible for bulk data migration, routine system updates, and platform database maintenance.

3. Problem-Solution Fit Matrix

Identified Pain Points	Proposed System Capability	Business Value & Outcome
Time-Consuming Manual Entry	Bulk Spreadsheet Ingestion	Automates bulk data loads from Excel/CSV files in seconds.
Format Mismatch & Field Errors	Automated & Custom Field Mapping	Ensures source spreadsheet columns accurately map to schema fields.
Duplicate Database Records	Coalesce Matching Engine	Checks unique keys to update existing records instead of duplicating.
Risk of Data Corruption	Intermediate Staging (Import Set Table)	Provides a isolated staging layer to validate data prior to commit.

4. Summary of Fit

The proposed solution directly eliminates manual data entry bottlenecks by combining automated file parsing, intermediate staging validation, and intelligent field mapping to ensure fast, reliable, and error-free database updates.